

28. Find the Index of the First Occurrence in a String:

Given two strings `needle` and `haystack`, return the index of the first occurrence of `needle` in `haystack`, or `-1` if `needle` is not part of `haystack`.

Example 1:

Input: `haystack = "sadbutsad"`, `needle = "sad"`
Output: `0`
Explanation: "sad" occurs at index 0 and 6.
The first occurrence is at index 0, so we return 0.

Example 2:

Input: `haystack = "leetcode"`, `needle = "leeto"`
Output: `-1`
Explanation: "leeto" did not occur in "leetcode", so we return -1.

Constraints:

- `1 <= haystack.length, needle.length <= 104`
- `haystack` and `needle` consists of only lowercase English characters.

Solution:

```
class Solution:
    def strStr(self, haystack: str, needle: str) -> int:
        return haystack.find(needle)
```

manual way

```

class Solution:
    def strStr(self, haystack: str, needle: str) -> int:
        if needle == "":
            return 0

        for i in range(len(haystack) - len(needle) + 1):
            if haystack[i:i+len(needle)] == needle:
                return i
        return -1

```

Two Pointer:

```

class Solution:
    def strStr(self, haystack: str, needle: str) -> int:
        if needle == "":
            return 0

        i = 0
        while i <= len(haystack) - len(needle):
            j = 0
            while j < len(needle) and haystack[i + j] == need
le[j]:
                j += 1

            if j == len(needle):
                return i

            i += 1

        return -1

```